

3N187

Silicon Dual Insulated-Gate Field-Effect Transistor

N-Channel Depletion Type

With Integrated Gate-Protection Circuits

For Military and Industrial Applications up to 300 MHz

RCA-3N187 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS⁴ pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ± 10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 3N187 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-3N187 make it useful for a wide variety of rf-amplifier applications at frequencies up to 300 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 3N187 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows

operation at maximum gain *without neutralization*; and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 3N187 is hermetically sealed in the metal JEDEC TO-72 package.

▲ Metal-Oxide-Semiconductor

Maximum Ratings,

Absolute-Maximum Values, at $T_A = 25^\circ\text{C}$

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	... -0.2 to +20	V
GATE No. 1-TO-SOURCE VOLTAGE, V_{G1S}	...	
Continuous (I_{d1})	... -6 to +3	V
Peak ac	... -6 to +6	V
GATE No. 2-TO-SOURCE VOLTAGE, V_{G2S}	...	
Continuous (I_{d2})	... -6 to 30% of V_{DS}	V
Peak ac	... -6 to +6	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	... +20	V
DRAIN CURRENT, I_D	... 50	mA
TRANSISTOR DISSIPATION P_T :	...	
At ambient	... up to 25°C	330 mW
temperatures	... above 25°C	derate linearly at 2.2 mW/ $^\circ\text{C}$
AMBIENT TEMPERATURE RANGE:	Storage and Operating	-65 to +175 $^\circ\text{C}$
LEAD TEMPERATURE (During Soldering):	At distance $\geq 1/32$ inch from seating surface for 10 seconds max.	265 $^\circ\text{C}$

* In accordance with JEDEC Registration Data Format JS-9 RDF-19A

Device Features

- Back-to-back diodes protect each gate against handling and in-circuit transients
- High forward transconductance — $g_{fs} = 12,000 \mu\text{mho}$ (typ.)
- High unneutralized RF power gain — $G_{ps} = 18 \text{ dB}$ (typ.) at 200 MHz
- Low VHF noise figure — 3.5 dB (typ.) at 200 MHz

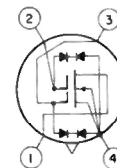
Applications

- RF amplifier, mixer, and IF amplifier in military, and industrial communications equipment
- Aircraft and marine vehicular receivers
- CATV and MATV equipment
- Telemetry and multiplex equipment

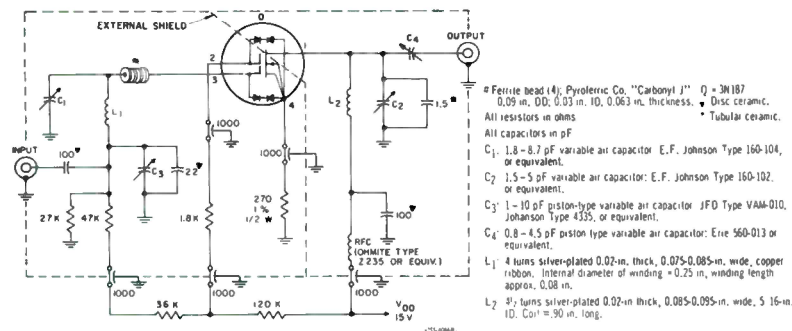
Performance Features

- Superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's
- Wide dynamic range permits large-signal handling before overload
- Virtually no agc power required
- Greatly reduces spurious responses in FM receivers

TERMINAL DIAGRAM



LEAD 1-DRAIN
LEAD 2-GATE No. 2
LEAD 3-GATE No. 1
LEAD 4-SOURCE, SUBSTRATE AND CASE



ELECTRICAL CHARACTERISTICS, at $T_A = 25^\circ\text{C}$ unless otherwise specified

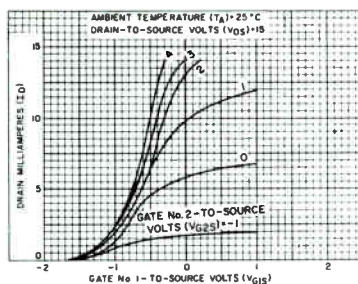
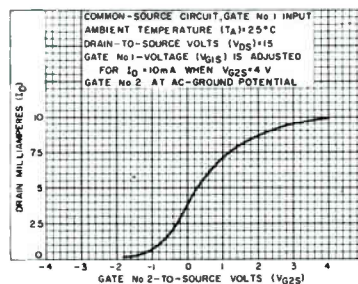
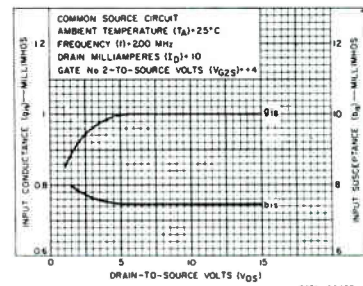
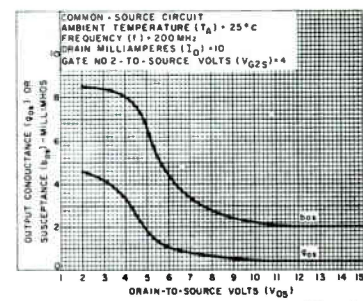
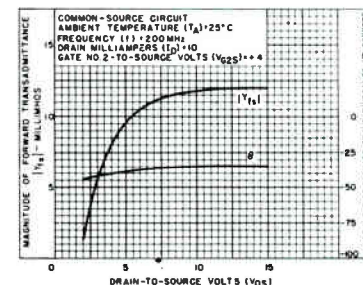
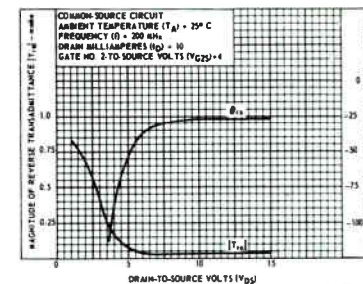
CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	$V_{DS} = +15\text{ V}, I_D = 50\text{ }\mu\text{A}$ $V_{G2S} = +4\text{ V}$	-0.5	-2	-4	V
Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	$V_{DS} = +15\text{ V}, I_D = 50\text{ }\mu\text{A}$ $V_{G1S} = 0$	-0.5	-2	-4	V
Gate No. 1-Terminal Forward Current	I_{G1SSF}	$V_{G1S} = +1\text{ V}$ $V_{G2S} = V_{DS} = 0$ $T_A = 25^\circ\text{C}$	-	-	50	nA
		$T_A = 100^\circ\text{C}$	-	-	5	μA
Gate No. 1-Terminal Reverse Current	I_{G1SSR}	$V_{G1S} = -6\text{ V}$ $V_{G2S} = V_{DS} = 0$ $T_A = 25^\circ\text{C}$	-	-	50	nA
		$T_A = 100^\circ\text{C}$	-	-	5	μA
Gate No. 2-Terminal Forward Current	I_{G2SSF}	$V_{G2S} = +6\text{ V}$ $V_{G1S} = V_{DS} = 0$ $T_A = 25^\circ\text{C}$	-	-	50	nA
		$T_A = 100^\circ\text{C}$	-	-	5	μA
Gate No. 2-Terminal Reverse Current	I_{G2SSR}	$V_{G2S} = -6\text{ V}$ $V_{G1S} = V_{DS} = 0$ $T_A = 25^\circ\text{C}$	-	-	50	nA
		$T_A = 100^\circ\text{C}$	-	-	5	μA
Zero-Bias Drain Current	I_{DS}	$V_{DS} = +15\text{ V}$ $V_{G2S} = +4\text{ V}$ $V_{G1S} = 0$	5	15	30	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	$V_{DS} = +15\text{ V}, I_D = 10\text{ mA}$ $V_{G2S} = +4\text{ V}, f = 1\text{ kHz}$	7000	12,000	18,000	μmho
Small-Signal, Short-Circuit Input Capacitance ¹	C_{iss}	$V_{DS} = +15\text{ V}, I_D = 10\text{ mA}$ $V_{G2S} = +4\text{ V}, f = 1\text{ MHz}$	4.0	6.0	8.5	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No. 1) ¹	C_{rss}		0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}		-	2.0	-	pF
Power Gain (see Fig. 1)	G_{PS}		16	18	22	dB
Maximum Available Power Gain	MAG		-	20	-	dB
Maximum Usable Power Gain (unneutralized)	MUG	$V_{DS} = +15\text{ V}, I_D = 10\text{ mA}$ $V_{G2S} = +4\text{ V}, f = 200\text{ MHz}$	-	20	-	dB
Noise Figure (see Fig. 1)	NF		-	3.5	4.5	dB
Magnitude of Forward Transmittance	$ Y_{fs} $		-	12,000	-	μmho
Phase Angle of Forward Transmittance	θ_{fs}		-	-35	-	Degrees
Magnitude of Reverse Transmittance	$ Y_{rs} $		-	25	-	μmho
Angle of Reverse Transmittance	θ_{rs}		-	-25	-	Degrees
Input Resistance	r_{iss}	$V_{DS} = +15\text{ V}, I_D = 10\text{ mA}$ $V_{G2S} = +4\text{ V}, f = 200\text{ MHz}$	-	1.0	-	k Ω
Output Resistance	r_{oss}		-	2.8	-	k Ω
Gate-to-Source Forward Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSF}$	$I_{G1SSF} = I_{G2SSF} = 100\text{ }\mu\text{A}$	6.5	10	-	V
Gate-to-Source Forward Breakdown Voltage: Gate No. 2	$V_{(BR)G2SSF}$		-	-	-	-
Gate-to-Source Reverse Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSR}$	$I_{G1SSR} = I_{G2SSR} = 100\text{ }\mu\text{A}$	-6.5	-10	-	V
Gate-to-Source Reverse Breakdown Voltage: Gate No. 2	$V_{(BR)G2SSR}$		-	-	-	-

¹ Limited only by practical design considerations.² Capacitance between Gate No. 1 and all other terminals³ Three-terminal measurement with Gate No. 2 and Source returned to ground terminal.

* In accordance with JEDEC Registration Data Format J5-9 R0F-19A

OPERATING CONSIDERATIONS

The flexible leads of the 3N187 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.

Fig. 6 - I_D vs. V_{G1S} Fig. 7 - I_D vs. V_{G2S} Fig. 8 - y_{is} vs. V_{DS} Fig. 9 - y_{os} vs. V_{DS} Fig. 10 - y_{fs} vs. V_{DS} Fig. 11 - y_{rs} vs. V_{DS}

3N187

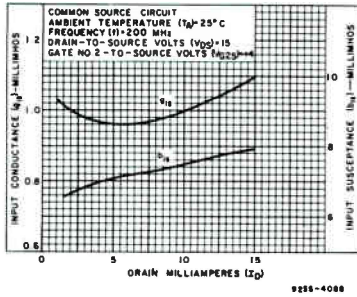


Fig. 12. y_{is} vs. I_D

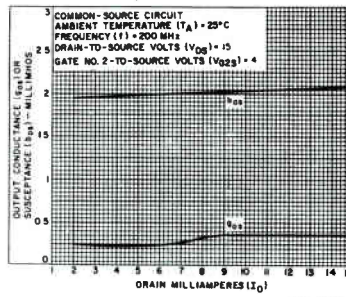


Fig. 13. y_{os} vs. I_D

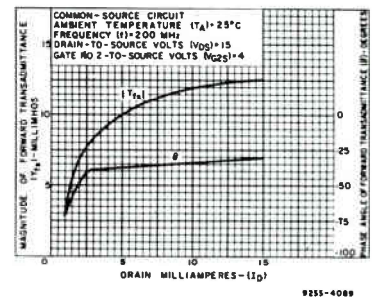


Fig. 14. y_{fs} vs. I_D

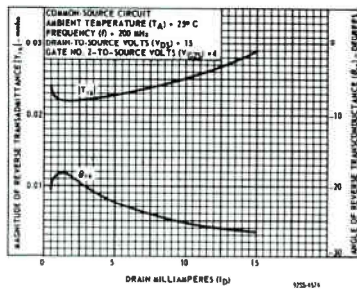


Fig. 15. y_{rs} vs. I_D

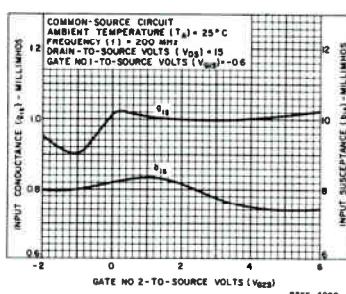


Fig. 16. y_{is} vs. V_{G2S}

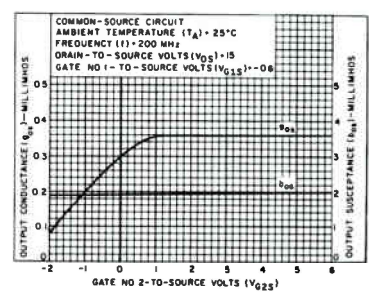


Fig. 17. y_{os} vs. V_{G2S}

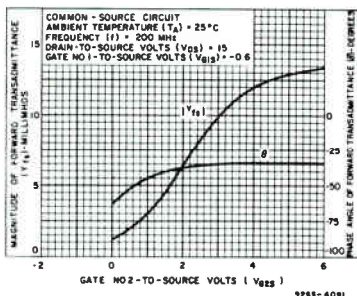


Fig. 18. y_{fs} vs. V_{G2S}

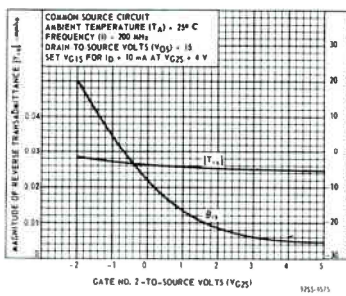


Fig. 19. y_{rs} vs. V_{G2S}

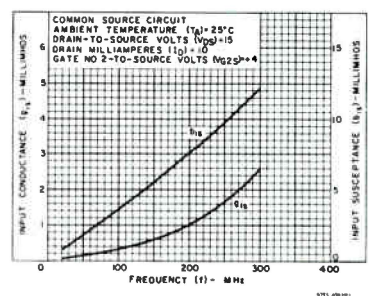


Fig. 20. y_{is} vs. frequency

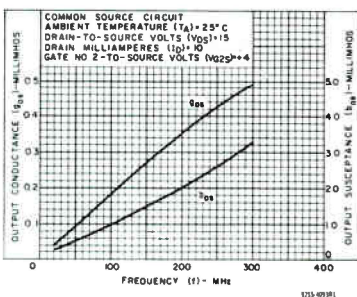


Fig. 21. y_{os} vs. frequency

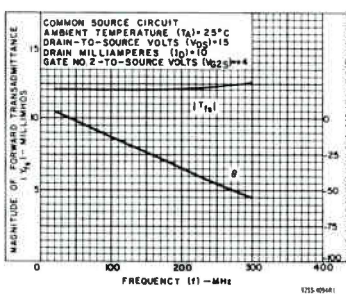


Fig. 22. y_{fs} vs. frequency

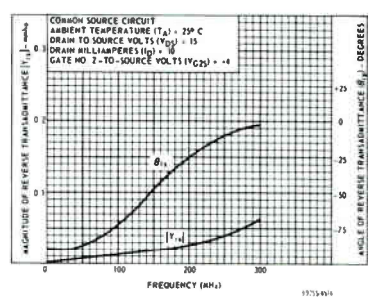


Fig. 23. y_{rs} vs. frequency

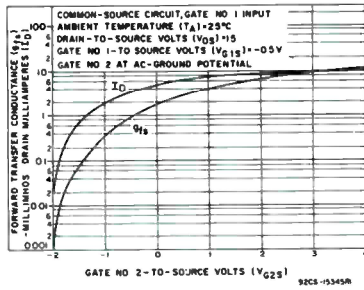


Fig. 24 - g_{fs} and I_D vs. V_{G2S}

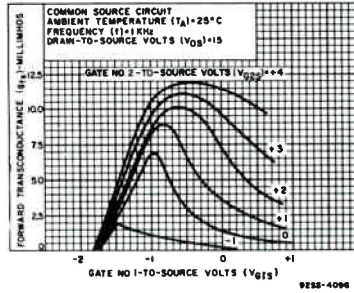


Fig. 25 - g_{fs} vs. V_{G1S}

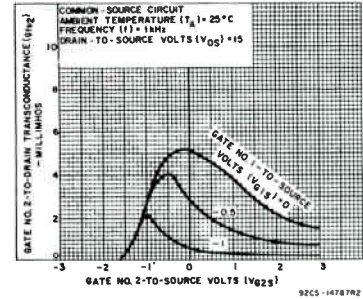


Fig. 26 - g_{fs2} vs. V_{G2S}